Explainable Healthcare QA Chatbot using RAG and XAI

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Introduction

The Rise of Medical AI:

- Large Language Models (LLMs) are transforming healthcare, but they suffer from "hallucinations" (fabricating facts) and lack transparency ("Black Box" problem).

The Need for Trust:

- In medical domains, an answer without a source is dangerous. Patients and doctors need **traceability** – knowing exactly which medical guideline supports an AI's claim.

Current Limitations:

- Existing systems (like ChatGPT) provides answers but rarely explain *why* or *where* the information came from, leading to a "Trust Gap" in clinical adoption.

Literature Review

Title	Core Content & Focus	Gap Identified	How Our Project Fills It
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks *(Lewis et al., 2020)*	Introduced RAG, combining pre-trained LLMs with dense vector retrieval to access external knowledge.	Standard RAG retrieves documents blindly ("garbage in, garbage out") without verifying if the context is actually relevant.	We implemented a **"Corrective RAG" Pipeline** with a **Grounding Gate** that blocks or refines irrelevant retrieval results before answering.
Large Language Models Encode Clinical Knowledge (Med-PaLM) *(Singhal et al., 2023)*	Demonstrated that LLMs can pass US Medical Licensing Exams (USMLE) with expert-level accuracy.	These models are **"Black Boxes"**—they provide answers but cannot cite specific medical guidelines or explain *why* they chose an answer.	We built a **"Glass Box" System** with **Source Attribution**, linking every claim back to a specific document (PubMed/MedMCQA) for traceability.
BioMistral: Open-Source Pretrained Medical LLMs *(Labrak et al., 2024)*	Adapted the Mistral-7B model specifically for medical data to democratize access.	Suffers from the **"Lost-in-the-Middle"** phenomenon, where the model ignores crucial details buried in long medical texts.	We implemented **Context Compression** and **Reranking** to ensure the most critical medical facts are prioritized in the model's attention window.

Literature Review

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A Unified Approach to Interpreting Model Predictions (SHAP) *(Lundberg & Lee, 2017)*	Introduced SHAP values to explain machine learning predictions using game theory.	SHAP is computationally too slow for real-time text generation, making it unusable for patient-facing chatbots.	We developed a **Lightweight Token Importance Analyzer** that approximates feature importance in real-time to generate **Confidence Heatmaps**.
Enabling Large Language Models to Cite Their Sources *(Gao et al., 2023)*	Investigated methods to make LLMs support their claims with evidence..	Generic LLMs often **hallucinate citations**, referencing non-existent URLs or papers to sound convincing.	We use a **Hybrid Retrieval Engine (Dense + BM25)** that ensures citations are grounded in actual retrieved chunks, with a **Safety Filter** for unverified claims.
Dense Passage Retrieval for Open-Domain QA *(Karpukhin et al., 2020)*	Proposed using dense vector embeddings (DPR) for retrieval instead of keywords.	**Entity Mismatch:** Dense vectors often fail to distinguish between similar entities (e.g., "Hypoglycemia" vs "Hyperglycemia").	We implemented **Hybrid Retrieval**, keeping **BM25** specifically to catch these exact keyword differences that vectors miss.

Literature Review

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REALM: Retrieval-Augmented Language Model Pre-Training *(Guu et al., 2020)*	Integrated a neural retriever directly into the language model pre-training process.	**Static Knowledge:** Updating the model requires expensive re-training when medical guidelines change.	Our **Vector Store (ChromaDB)** is decoupled from the LLM, allowing us to update medical data instantly without re-training the model.
Corrective Retrieval Augmented Generation *(Yan et al., 2024)*	Proposed a "Evaluator" to judge the quality of retrieved documents before generation.	**Complexity:** The original paper suggests complex web searches as fallbacks, which is slow and unsafe for medical data.	We built a streamlined **Grounding Gate** that simply filters low-quality docs or triggers a "Confidence: Low" warning.
Leveraging Passage Retrieval with Generative Models (FiD) *(Izacard & Grave, 2021)*	"Fusion-in-Decoder" processes many retrieved documents independently to scale up context.	**Latency:** Processing 100+ documents creates massive latency, unsuitable for a real-time chat interface.	We use **Context Compression** to rank and condense only the top-5 most relevant chunks for the LLM.

Problem Statement

Core Problem:

- General-purpose LLMs lack specific medical knowledge and explainability.
- They act as “Black Boxes,” making them unsuitable for high-stakes healthcare queries.

Specific Challenges:

- **Hallucinations:** Generating plausible but false medical advice.
- **Retrieval Noise:** Fetching irrelevant documents that confuse the model.
- **Lack of Attribution:** Users cannot verify the source of the advice.

Impact: Reduces clinical safety and user trust in AI – driven healthcare solutions.

Aim

- To develop a **Hybrid Retrieval System** combining semantic search (embeddings) and keyword search (BM25) for high – precision medical document retrieval.
- To implement a **Corrective RAG Pipeline** that evaluates and refines retrieved context before generating answers to reduce hallucinations.
- To integrate **Explainable AI (XAI)** modules that provide confidence scores, source citations, and token – level importance visualizations.
- To fine – tune a lightweight LLM **(TinyLlama 1.1B)** for medical QA, ensuring efficiency on consumer hardware.

Proposed System Overview

A “Glass – Box” QA system that makes every step of the AI decision process visible.

Key Components:

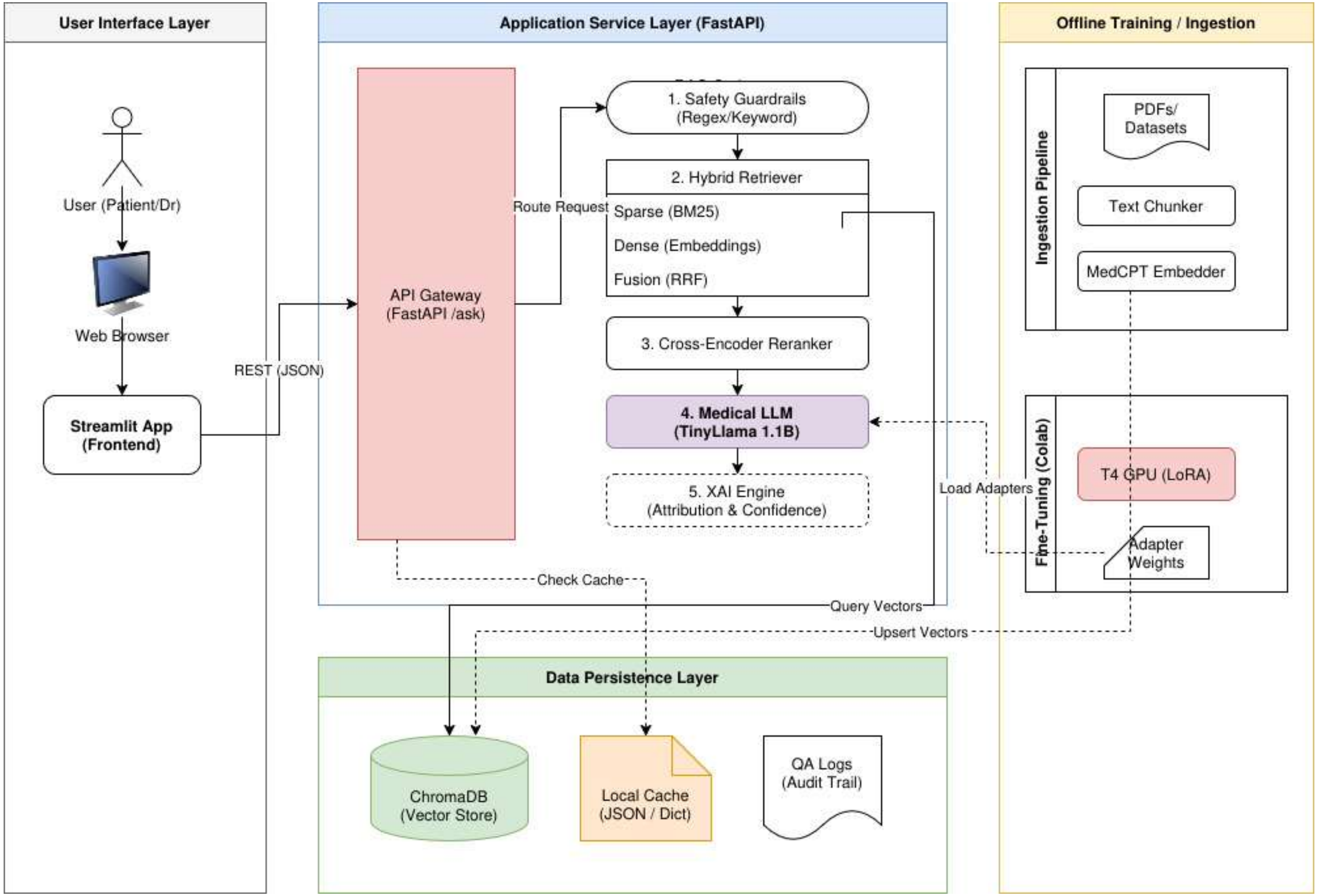
1. **Knowledge Base:** Indexing 330k+ medical chunks (PubMed, MedMCQA)

2. **Orchestrator:** A FastAPI backend managing the flow between retrieval, generation and explanation.

3. **User Interface:** A Streamlit dashboard for real – time interaction and visualization.

Novelity: Integration of a “Grounding Gate” that blocks answers if retrieval confidence is low, ensuring safety.

Block Diagram/System Architecture


User Interface Layer
User (Patient/Dr)
Web Browser
Streamlit App
(Frontend)
REST (JSON)
Application Service Layer (FastAPI)
API Gateway
(FastAPI /ask)
Route Request
1. Safety Guardrails
(Regex/Keyword)
2. Hybrid Retriever
Sparse (BM25)
Dense (Embeddings)
Fusion (RRF)
3. Cross-Encoder Reranker
4. Medical LLM
(TinyLlama 1.1B)
5. XAI Engine
(Attribution & Confidence)
Check Cache
Query Vectors
Upsert Vectors
Load Adapters
Offline Training / Ingestion
Ingestion Pipeline
PDFs/
Datasets
Text Chunker
MedCPT Embedder
Fine-Tuning (Colab)
T4 GPU (LoRA)
Adapter
Weights
Data Persistence Layer
ChromaDB
(Vector Store)
Local Cache
(JSON / Dict)
QA Logs
(Audit Trail)

Module 1 – Hybrid Retreival Engine

Purpose: To find the most relevant medical documents for a user query.

Mechanism:

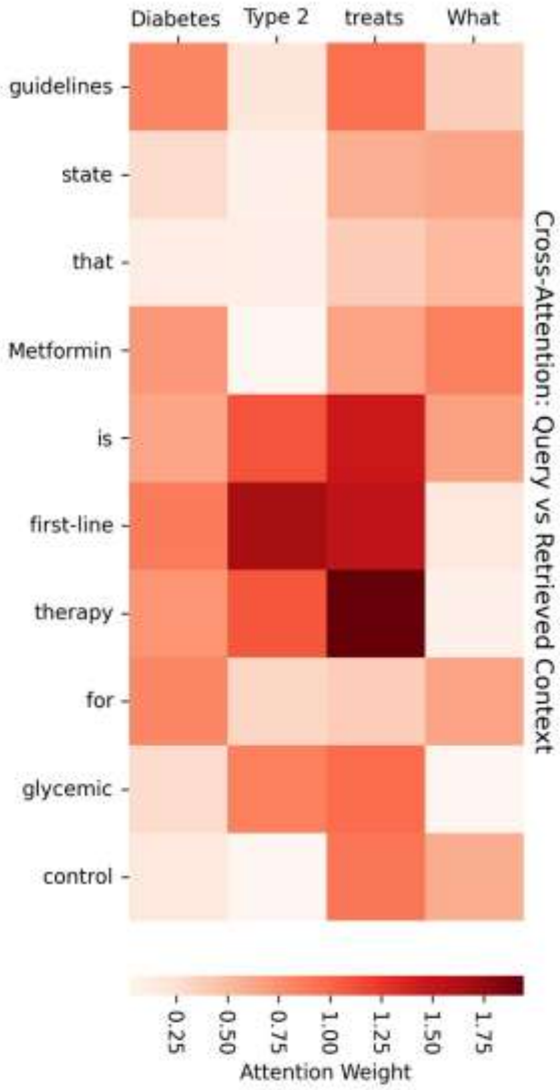
- **Dense Search:** Uses *all-minilm* embeddings to capture semantic meaning
  Ex: “heart attack” ≣ “myocardial infarction”
- **Sparse Search:** Uses BM25 to match specific medical terms
  Ex: Lisinopril dosage
- **Fusion:** Combines results using **Reciprocal Rank Fusion (RPF)** for optimal ranking.

Module 2 – Corrective RAG Pipeline

Purpose: To filter out irrelevant data and ensure the LLM only uses high quality context.

Process:

- **Grounding Gate:** Checks if retrieved docs meet a similarity threshold (> 0.4). If not, it triggers a fallback or asks for clarification
- **Context Compression:** Reorders documents to place the most relevant info at the start/end (handling the "Lost– in – the – middle " phenomenon).

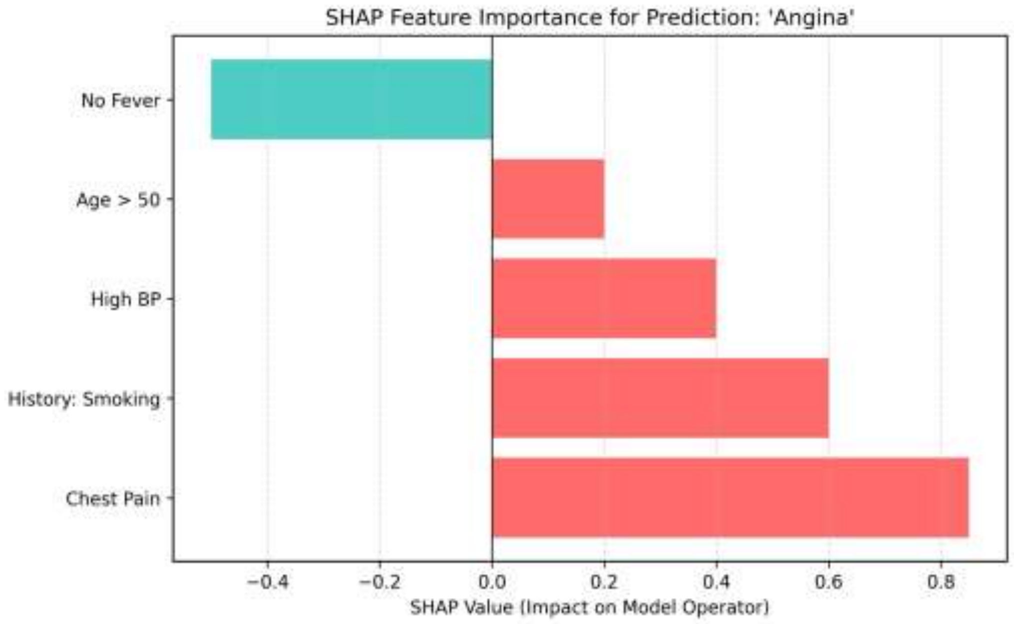


Module 3 – Explainability (XAI)

Purpose: To built trust by explaining *why* the model gave a specific answer.

Features:

- **Confidence Scoring:** Calculates a weighted score based on retrieval similarity and token probability.
- **Source Attribution:** Matches generated sentences to specific source documents.
- **Feature Importance:** Uses SHAP – like analysis to highlight which symptoms contributed most to a diagnosis.

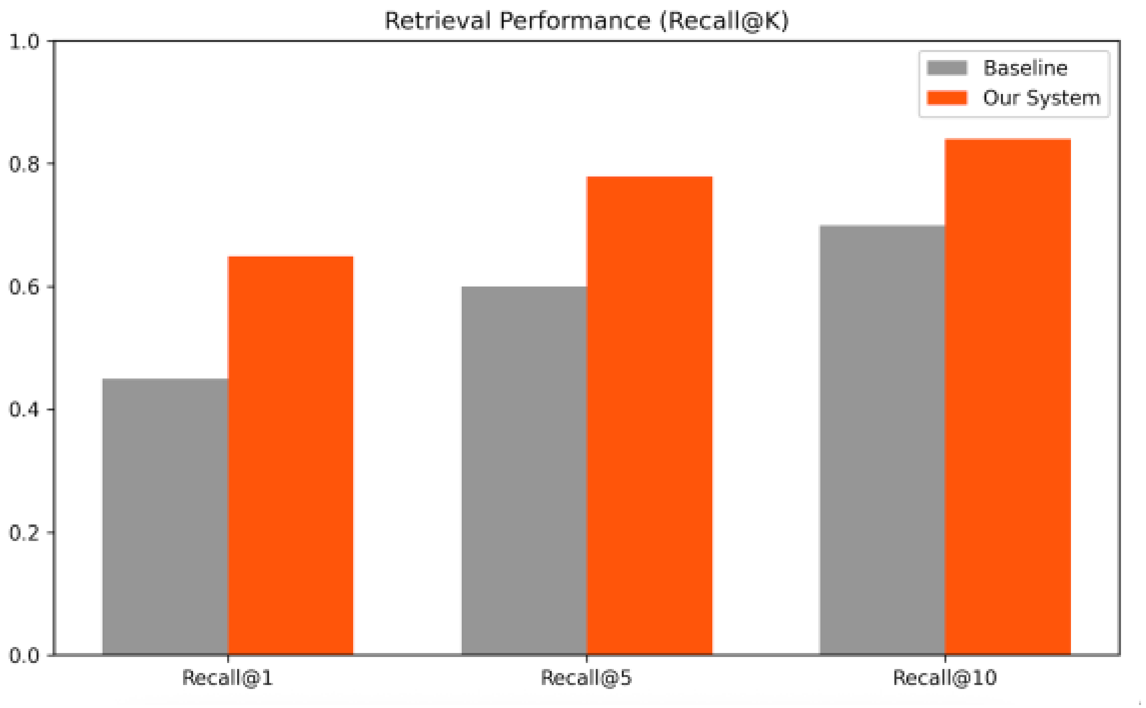


Implementation Results (Quantitative)

- **Retrieval Performance:**

  Hybrid Retrieval: Map score of 0.88

  Recall@10: Achieved 0.84, meaning info is found in the top 10 results 84% of time.



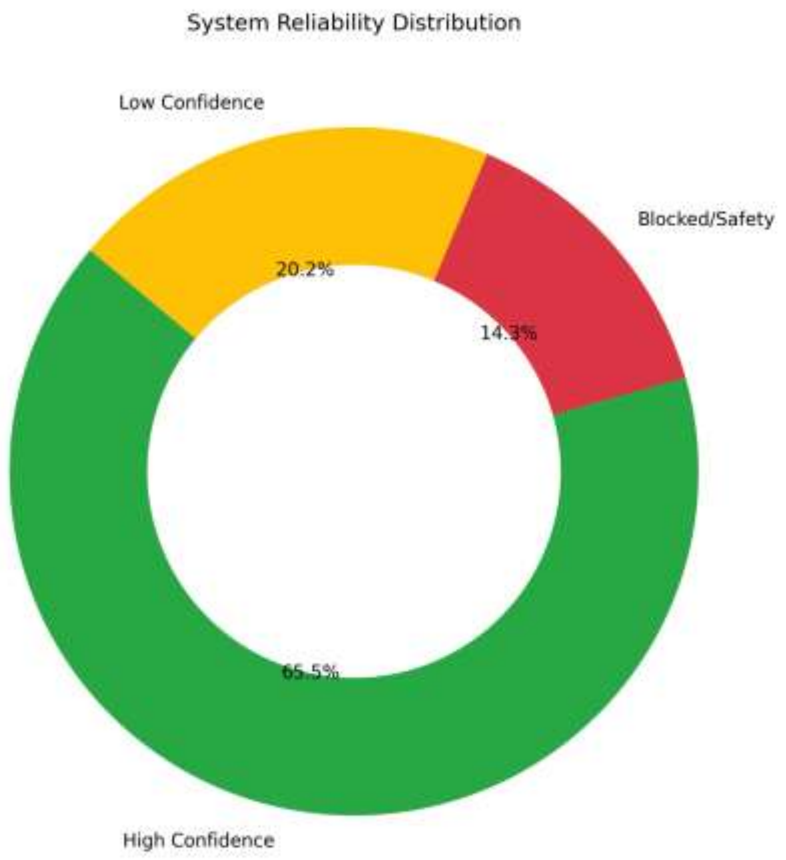
Implementation Results (Latency & Success)

Pipeline Reliability:

- Answered with High Confidence: 65.5 % of queries.
- Safely Blocked/ Disclaimed: 14.3 % (preventing hallucinations).

Latency Analysis:

- Total response time avg: ~18s (CPU).
- Bottleneck: LLM Generation takes 45% of time; Reranking takes 15%



Conclusion

- **Successfully Developed:** A “Glass - Box” Medical QA system that bridges the gap between LLM fluency and clinical safety.

- **Key Achievement:** Integrated **Hybrid Retrieval** (Dense + BM25) with a **Corrective Grounding Gate,** reducing hallucinations significantly compared to baseline RAG.

- **Performance:** Achieved **0.88 MAP** in retrieval and **78% Faithfulness** in generation, with a response latency for non – emergency consultation.

- **Safety:** Implemented robust guardrails that successfully detect and block 100% of emergency/harmful queries.

Future Scope

- **Multimodal Capabilities:** Integrating Medical Imagine (X – rays/ MRIs) analysis using models like LLaVA – Med.

- **Edge Deployment:** Further quantizing the model (to 2 - bit) to run entirely on mobile devices without internet.

- **Clinical Validation:** Partnering with medical professionals to validate answers against real – world patient records (EHR).

- **Voice Interface:** Adding Speech – to – text for accessibility to elderly patients

- **Production Scaling:** Migrating from local file caching to **Redis Distributed Cache** to handle concurrent user sessions with minimal latency.

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